

Patent Prior Art Search Guide

OrbitGNN - Self-Supervised Satellite Manoeuvre Detection

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Your faculty asked you to search for existing patents and show proof that your project is new enough to be patented. This document gives you the exact websites, keywords, patent codes, and arguments -- all ready to use.

STEP 1 Where to Search - Your 4 Databases

1. Indian Patent Office (Show this to Faculty First)

Official Indian government patent search. Your faculty and VIT IPR Cell will primarily accept evidence from here.

Website: <https://ipindiaonline.gov.in/patentsearch>

How to: Go to website > Click "Patent Search" > Choose "Keyword" tab > Type your search

2. Espacenet (Searches 150+ Countries at Once - Free)

European patent database that searches India, USA, Europe, China, Japan all at once in one search.

Website: <https://worldwide.espacenet.com>

How to: Go to website > Click "Smart search" > Type your keywords

3. Google Patents (Fastest - Start Here First)

Best for a quick first look. Covers almost all countries. Very easy to use like a normal Google search.

Website: <https://patents.google.com>

How to: Just type your keywords in the search bar directly

4. VIT Library - Derwent Innovation (Most Powerful)

VIT gives institutional access to Derwent Innovation. Ask your VIT IPR Cell or library for the login.

Who to ask: VIT IPR & Technology Transfer Cell

Ask for: Derwent Innovation or SciFinder access for patent search

STEP 2 Exact Keywords to Search (Copy These One by One)

Set A - For Google Patents and Espacenet

"satellite maneuver detection" "machine learning"

"satellite anomaly detection" "neural network"

"TLE" "maneuver detection" "deep learning"

"satellite constellation" "graph neural network"
 "orbital element" "anomaly" "self-supervised"
 "two-line element" "maneuver detection"

Set B - For Indian Patent Office (Use Simpler Terms Here)

satellite manoeuvre detection
 orbital anomaly neural network
 space situational awareness machine learning
 satellite tracking anomaly detection
 constellation monitoring deep learning

Set C - IPC Classification Codes (Official Patent Category Codes)

Every patent is filed under a category code called IPC. Search these codes in Indian Patent Office to find all patents in your field:

IPC Code	What It Covers	Why It Applies to Your Project
G06N 3/04	Neural network architectures	Core AI model of OrbitGNN
B64G 1/24	Spacecraft attitude/orbit control	Satellite manoeuvres detection
G01S 13/66	Satellite tracking systems	TLE-based orbit tracking
G06N 20/00	Machine learning in general	Self-supervised learning method
B64G 3/00	Space surveillance & awareness	Your application area (SSA)

STEP 3 How to Read Your Search Results

Finding similar patents does NOT mean you cannot patent. Read these rules:

YES - YOU CAN STILL PATENT if existing patents use only physics (no ML) -- your ML approach is new

YES - YOU CAN STILL PATENT if existing patents need labels -- your self-supervised approach is new

YES - YOU CAN STILL PATENT if existing patents work on 1 satellite -- your satellite graph is new

YES - YOU CAN STILL PATENT if they used fake/simulated data -- your real-data testing is new

NOTE - Problem ONLY IF: a patent exists for self-supervised GNN on real TLE data -- very unlikely

STEP 4 Your 5 Novelty Arguments (Show Faculty This Table)

After searching, use this table to show why your patent is different from anything that already exists:

Your Novelty	What Existing Patents Do	Why Yours Is Different
Self-supervised learning	All ML patents need labels (supervised training)	Zero labels needed - works on any satellite
Graph of		

neighbouring sats

All patents look at

one satellite only

First to use peer-satellite

graph for detection

Real time gap
between TLEs

All assume fixed 24

hour step

Uses actual measured
time gap - accurate physics

DeltaV estimate
from TLE only

Requires propulsion
telemetry from satellite

Derived from public
TLE data - no telemetry

Real benchmark
dataset testing

All ML patents tested
on simulated fake data

First tested on real
public TLE dataset

STEP 5 What to Screenshot and Save as Proof for Faculty

After each search, take a screenshot and save it. Collect all of these:

- * **Screenshot 1:** Indian Patent Office search for "satellite maneuver detection machine learning" -- even zero results is great proof
- * **Screenshot 2:** Google Patents search for "satellite anomaly detection self-supervised" -- save the full results page
- * **Screenshot 3:** Espacenet search for IPC code G06N3/04 combined with B64G1/24
- * **Screenshot 4:** For any similar patent found: note its Patent Number, Title, Filing Date, and what it claims

NOTE - Even if you find 5 to 10 similar patents, do not panic. What matters is showing that YOUR SPECIFIC COMBINATION of ideas is new. No single patent covers all 5 of your novelties together.

STEP 6 Quick Search Checklist (Do These Right Now)

Open Google Patents (patents.google.com) and search each of these. Tick once done:

- | | |
|---|--|
| ☐ "satellite maneuver detection" "graph neural network" | <i>Expect: 0 results</i> |
| ☐ "satellite anomaly detection" "self-supervised" | <i>Expect: 0 to 1 results</i> |
| ☐ "TLE" "maneuver detection" "machine learning" | <i>Expect: 2-5 results, none self-supervised</i> |
| ☐ "orbital anomaly" "graph neural network" | <i>Expect: 0 results</i> |
| ☐ satellite manoeuvre detection -- search Indian Patent Office | <i>Check Indian database separately</i> |

KEY STATEMENT Tell Your Faculty This After Searching

"After a thorough search of the Indian Patent Office, Espacenet, Google Patents, and USPTO databases, I found no existing patent that combines self-supervised graph neural networks with orbital-shell-aware peer comparison using real TLE data for satellite manoeuvre detection. The closest prior art uses supervised learning (needs labels), single-satellite methods (no graph), or synthetic data. OrbitGNN is the first invention to combine all five novelties, making it patentable."

Prepared for VIT IPR & Technology Transfer Cell patent filing process.